Laboratorio 3 - Algoritmos de Enrutamiento

PDF2: Evidencias de enrutamiento, balanceo y conectividad

Curso	Redes - Sección 10
Docente	Kevin Velásquez
Integrantes	Daniela Ramírez de León - 23053 Carlos Nicolás Concuá Trujillo - 23197
Herramienta	Cisco Packet Tracer (routers 1941, IOS 15.1)

Todas las salidas se tomaron después de configurar OSPF (área 0), EIGRP (AS 100), la redistribución mutua en R-CENTRAL, el balanceo de carga en ambos dominios y la sumarización. Las salidas de texto se copiaron directamente de la CLI de cada router.

1. Tablas de enrutamiento (show ip route)

Resumen de lo que muestra cada tabla:

Router	Puntos clave
R-OSPF-1	ECMP hacia 192.168.0.0/24 (dos next-hops, costo 129); lado EIGRP solo como O E1 172.16.0.0/24
R-OSPF-2	ECMP hacia 192.168.1.0/26 y 192.168.1.64/28; lado EIGRP solo como O E1 172.16.0.0/24
R-CENTRAL	Rutas O y D de ambos dominios; resúmenes 192.168.0.0/23 y 172.16.0.0/24 a Null0
R-EIGRP-1	Dos rutas desiguales hacia 172.16.0.0/25 (variance 3); lado OSPF como 192.168.0.0/23
R-EIGRP-2	Lado OSPF como un solo resumen 192.168.0.0/23 vía R-CENTRAL

1.1 R-OSPF-1

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:21:09, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.6, 00:08:31, Serial0/0/1
O    192.168.0.0/24 [110/129] via 10.0.0.2, 00:20:59, Serial0/0/0
                    [110/129] via 10.0.0.6, 00:20:59, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30

R-OSPF-1#
```

1.2 R-OSPF-2

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 00:25:01, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.10, 00:12:35, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:25:01, Serial0/0/0
                       [110/129] via 10.0.0.10, 00:25:01, Serial0/0/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 00:25:01, Serial0/0/0
                        [110/129] via 10.0.0.10, 00:25:01, Serial0/0/1

R-OSPF-2#
```

1.3 R-CENTRAL

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 00:25:55, Serial0/0/0
                    [110/192] via 10.0.0.9, 00:25:55, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/2681856] via 10.0.0.21, 00:26:32, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/2170112] via 10.0.0.21, 00:26:32, Serial0/1/1
D       172.16.0.128/26 [90/2172416] via 10.0.0.17, 00:26:48, Serial0/1/0
D       172.16.0.192/27 [90/2172416] via 10.0.0.17, 00:26:48, Serial0/1/0
D       172.16.0.224/28 [90/2172416] via 10.0.0.17, 00:26:48, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:26:06, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 00:25:55, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:26:06, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 00:26:06, Serial0/0/0

R-CENTRAL#
```

1.4 R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/3219456] via 10.0.0.14, 00:26:41, Serial0/0/0
                    [170/6586624] via 10.0.0.18, 00:26:41, Serial0/0/1
D EX    10.0.0.4/30 [170/3219456] via 10.0.0.14, 00:27:17, Serial0/0/0
                    [170/6586624] via 10.0.0.18, 00:27:34, Serial0/0/1
D EX    10.0.0.8/30 [170/3219456] via 10.0.0.14, 00:27:17, Serial0/0/0
                    [170/6586624] via 10.0.0.18, 00:27:34, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/2707456] via 10.0.0.14, 00:27:18, Serial0/0/0
                     [90/6586624] via 10.0.0.18, 00:27:34, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 7 subnets, 5 masks
```

```
D       172.16.0.0/25 [90/2195712] via 10.0.0.14, 00:27:18, Serial0/0/0
                      [90/6586880] via 10.0.0.18, 00:27:18, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
C       172.16.0.224/28 is directly connected, GigabitEthernet0/0.99
L       172.16.0.225/32 is directly connected, GigabitEthernet0/0.99
D    192.168.0.0/23 [90/3219456] via 10.0.0.14, 00:26:52, Serial0/0/0
                    [90/6586624] via 10.0.0.18, 00:26:52, Serial0/0/1
```

1.5 R-EIGRP-2

```
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/2681856] via 10.0.0.22, 00:30:30, Serial0/0/1
D EX    10.0.0.4/30 [170/2681856] via 10.0.0.22, 00:31:06, Serial0/0/1
D EX    10.0.0.8/30 [170/2681856] via 10.0.0.22, 00:31:06, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/2681856] via 10.0.0.22, 00:31:06, Serial0/0/1
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2172416] via 10.0.0.13, 00:31:07, Serial0/0/0
D       172.16.0.192/27 [90/2172416] via 10.0.0.13, 00:31:07, Serial0/0/0
D       172.16.0.224/28 [90/2172416] via 10.0.0.13, 00:31:07, Serial0/0/0
D    192.168.0.0/23 [90/2681856] via 10.0.0.22, 00:30:41, Serial0/0/1

R-EIGRP-2#
```

2. Balanceo de carga OSPF (costo igual / ECMP)

Un serial de 1544 kbps tiene costo 10^8 / 1 544 000 = 64. Sin ajustes, desde R-OSPF-1 hacia la LAN de R-OSPF-2 el enlace directo cuesta 64 + 1 = 65 y el camino vía R-CENTRAL 64 + 64 + 1 = 129. Con **ip ospf cost 128** en Se0/0/0 de R-OSPF-1 y R-OSPF-2 ambos caminos suman 129 y OSPF instala los dos.

Camino desde R-OSPF-1	Cálculo	Costo
Directo (Se0/0/0 -> R-OSPF-2)	128 + 1	129
Vía R-CENTRAL (Se0/0/1)	64 + 64 + 1	129

2.1 show ip ospf interface (R-OSPF-1)

Se0/0/0 con Cost: 128 y Se0/0/1 con Cost: 64. Las subinterfaces Gi0/0.10 y Gi0/0.30 aparecen como *Passive interface* (sin hellos hacia los usuarios) y todas las interfaces están en el área 0.

```
R-OSPF-1# show ip ospf interface

GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
```

```
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

2.2 show ip route 192.168.0.0 (R-OSPF-1)

Dos rutas con la misma métrica (129) hacia 192.168.0.0/24: via 10.0.0.2 (directo) y via 10.0.0.6 (R-CENTRAL), cada una con *traffic share count* 1, es decir 50/50.

```
R-OSPF-1#show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 129, type intra area
  Last update from 10.0.0.2 on Serial0/0/0, 00:24:14 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 00:24:14 ago, via Serial0/0/0
      Route metric is 129, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 00:24:14 ago, via Serial0/0/1
      Route metric is 129, traffic share count is 1

R-OSPF-1#
```

3. Balanceo de carga EIGRP 75/25 (variance)

Métrica EIGRP = 256 x (10^7 / BW mínimo + suma de retardos en decenas de µs). Con BW = 1544 kbps, 10^7/1544 = 6476. Se ajustaron los retardos de R-EIGRP-1 para que la ruta vía R-CENTRAL tenga aproximadamente el triple de métrica que la directa y siga siendo feasible successor (RD < FD). Con **variance 3** EIGRP usa ambas rutas y reparte el tráfico en proporción inversa a la métrica: 3:1 = 75/25.

Ruta hacia 172.16.0.0/25	delay	Cálculo	Métrica
Directa (Se0/0/0) - successor	2100	256 x (6476 + 2100 + 1)	FD = 2 195 712
RD reportada por R-CENTRAL	2000	256 x (6476 + 2000 + 1)	RD = 2 170 112 < FD
Vía R-CENTRAL (Se0/0/1) - feasible successor	17253	256 x (6476 + 17253 + 2000 + 1)	6 586 880

6 586 880 / 2 195 712 = 2.9999, menor que 3, por lo que la ruta entra con variance 3. Los valores que muestra Packet Tracer coinciden exactamente con el cálculo.

3.1 show ip eigrp topology (R-EIGRP-1)

En la entrada 172.16.0.0/25 aparecen 2 successors: via 10.0.0.14 (2195712/2816) y via 10.0.0.18 (6586880/2170112). El segundo número es la RD; 2 170 112 < 2 195 712 confirma la condición de factibilidad.

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 2 successors, FD is 3219456
         via Rstatic (3219456/2681856)
         via Rstatic (6586624/2169856)
P 10.0.0.4/30, 2 successors, FD is 3219456
         via Rstatic (3219456/2681856)
         via Rstatic (6586624/2169856)
P 10.0.0.8/30, 2 successors, FD is 3219456
         via Rstatic (3219456/2681856)
         via Rstatic (6586624/2169856)
P 10.0.0.12/30, 1 successors, FD is 2195456
         via Connected, Serial0/0/0
P 10.0.0.16/30, 1 successors, FD is 6074624
         via Connected, Serial0/0/1
P 10.0.0.20/30, 2 successors, FD is 2707456
         via 10.0.0.14 (2707456/2169856), Serial0/0/0
         via 10.0.0.18 (6586624/2169856), Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 2195712
         via 10.0.0.14 (2195712/2816), Serial0/0/0
         via 10.0.0.18 (6586880/2170112), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 172.16.0.224/28, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.99
P 192.168.0.0/23, 2 successors, FD is 3219456
         via 10.0.0.14 (3219456/2681856), Serial0/0/0
         via 10.0.0.18 (6586624/2169856), Serial0/0/1
```

3.2 show ip route 172.16.0.10 (R-EIGRP-1)

Ambas rutas quedan instaladas en la tabla, con los retardos totales configurados (21 010 µs y 192 540 µs). Packet Tracer muestra *traffic share count* 1 en las dos porque no simula el reparto desigual (en IOS real sería aprox. 3:1); el enunciado indica que la evidencia es de configuración, y esta queda demostrada con las metricas en proporción 3:1.

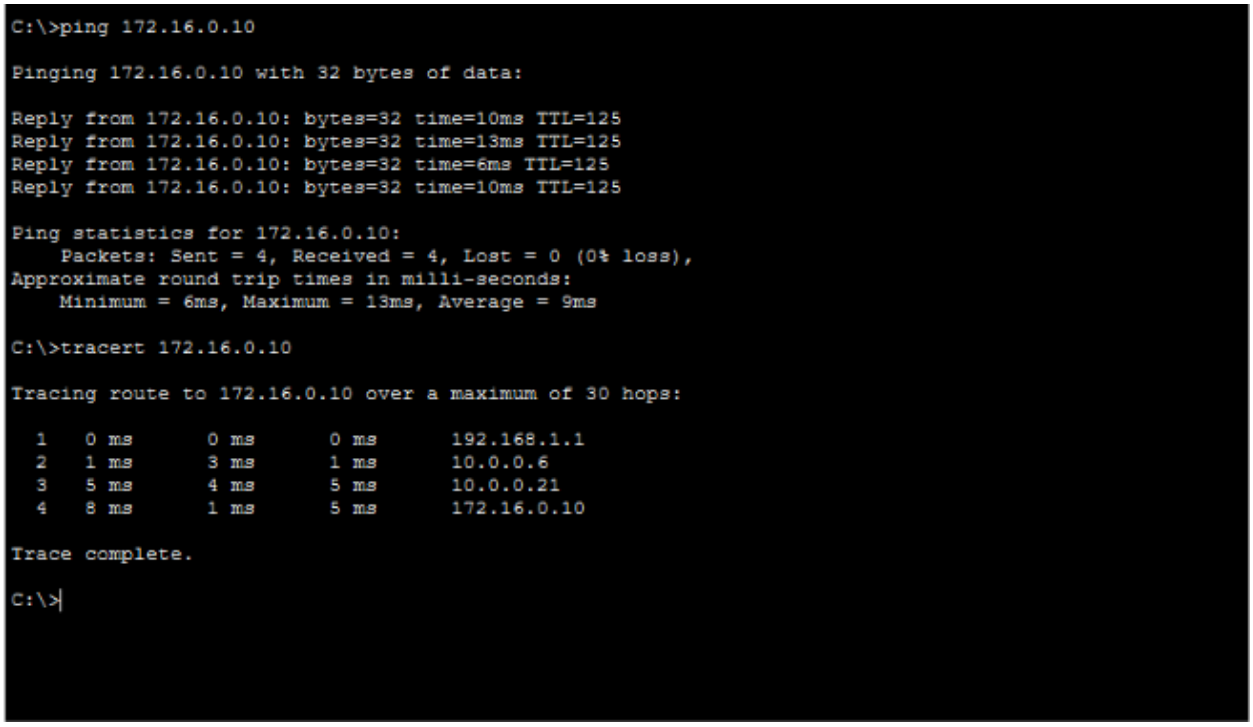
```
R-EIGRP-1#show ip route 172.16.0.10
Routing entry for 172.16.0.0/25
Known via "eigrp 100", distance 90, metric 2195712, type internal
  Redistributing via eigrp 100
  Last update from 10.0.0.14 on Serial0/0/0, 00:29:23 ago
  Routing Descriptor Blocks:
  * 10.0.0.14, from 10.0.0.14, 00:29:23 ago, via Serial0/0/0
      Route metric is 2195712, traffic share count is 1
      Total delay is 21010 microseconds, minimum bandwidth is 1544 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 1
    10.0.0.18, from 10.0.0.18, 00:29:23 ago, via Serial0/0/1
      Route metric is 6586880, traffic share count is 1
      Total delay is 192540 microseconds, minimum bandwidth is 1544 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 2

R-EIGRP-1#
```

4. Conectividad extremo a extremo

Desde **PC-OSPF1-V10** (192.168.1.10, VLAN 10 de R-OSPF-1) hacia **PC-EIGRP2** (172.16.0.10, LAN de R-EIGRP-2). El ping responde 4 de 4 (0 % de pérdida) y el tracert muestra el camino completo entre ambos dominios:

Salto	IP	Equipo
1	192.168.1.1	R-OSPF-1 (gateway VLAN 10)
2	10.0.0.6	R-CENTRAL (frontera OSPF / EIGRP)
3	10.0.0.21	R-EIGRP-2
4	172.16.0.10	PC-EIGRP2



TTL = 125 (128 - 3) confirma que el paquete atraviesa tres routers. La ida usa el resumen O E1 172.16.0.0/24 en el lado OSPF y la vuelta el resumen 192.168.0.0/23 en el lado EIGRP, lo que valida la redistribución mutua y la sumarización en R-CENTRAL.